

MALLA REDDY TECHNICAL CAMPUS

A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH

Deemed to be University

Approved by UGC & AICTE New Delhi

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COLLEGE EXAM SEATING ALLOCATION

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College Exam Room Allocation

- **Efficient Allocation of Examination Rooms**
- University examinations involve multiple subjects and large numbers of students.
- Each subject may have a different number of students.
- Examination rooms have different seating capacities.
- The goal is to allocate students to rooms efficiently while satisfying all constraints.
- **Presented by:**
Name: _____
Roll No.: _____
Course: _____

Problem Statement

- A university needs to schedule examinations for multiple subjects using a limited number of available rooms.
- **Given:**
- A set of examination subjects.
- Number of students registered for each subject.
- A set of rooms.
- Capacity of each room.
- **Objective:**
- Assign students to rooms so that:
- Every student gets a seat.
- Room capacity is not exceeded.
- The number of rooms used is minimized where possible.
- Room usage is efficient.
- No student is assigned to two rooms for the same examination.

Example Input

- Suppose the university has **5 subjects** and **6 examination rooms**.

Subject	Students
Mathematics	120
Physics	85
Chemistry	100
Computer Science	60
English	45

Available Rooms

Room	Capacity
R1	100
R2	80
R3	60
R4	50
R5	40
R6	30

The allocation system must divide students among rooms without exceeding room capacities.

Major Constraints

- The allocation strategy must consider several constraints.
- **Hard Constraints**
- Students assigned to a room must not exceed its capacity.
- Every registered student must receive a seat.
- A room cannot host more students than its physical capacity.
- A student should not be assigned to overlapping examinations.
- A room should not be assigned to two subjects at the same time.
- **Soft Constraints**
- Minimize unused seats.
- Minimize the number of rooms used.
- Keep allocations simple and easy to manage.
- Prefer balanced room utilization.

Proposed Allocation Strategy

- A capacity-based greedy allocation can be used.
- Basic idea:
- Sort subjects by number of students, from largest to smallest.
- Sort available rooms by capacity, from largest to smallest.
- Assign the largest suitable room to the subject.
- If one room is insufficient, use additional rooms.
- Continue until all students are accommodated.
- Calculate unused seats and room utilization.

Allocation Algorithm

- Step-by-step algorithm
- **Input:** Subjects, student counts, rooms, room capacities.
- **Algorithm:**
 - Sort subjects in descending order of student count.
 - Sort rooms in descending order of capacity.
 - Select the largest remaining subject.
 - Select a suitable available room.
 - Assign students until the room is full or all students are accommodated.
 - If students remain, assign another suitable room.
 - Repeat until the subject is completely allocated.
 - Continue with the next subject.
 - Calculate total seats used and unused seats.

Pseudocode

- sort subjects by student count descending
- sort rooms by capacity descending
- for each subject:
 - remaining = subject.students
 - while remaining > 0:
 - choose suitable available room
 - allocated = min(remaining, room.capacity)
 - assign allocated students to room
 - remaining = remaining - allocated

Sample Allocation

Mathematics — 120 students

R1 = 100 students

R3 = 20 students

Chemistry — 100 students

R2 = 80 students

R4 = 20 students

Physics — 85 students

R3 = 60 students

R4 = 25 students

Computer Science — 60 students

R3 = 60 students

English — 45 students

R4 = 45 students

Note: In an actual examination timetable, the same room can be reused by different subjects if their examinations occur at different times.

Room Utilization

- Room utilization measures how effectively the available seating capacity is being used.
- Formula
- Room Utilization (%)
- $$\text{Utilization} = \frac{\text{Students Allocated}}{\text{Room Capacity}} \times 100$$

Example

If 45 students are assigned to a room with 50 seats:

$$\frac{45}{50} \times 100 = 90\%$$

Therefore, the room is **90% utilized**.

Benefits

Identifies wasted capacity.

Helps reduce the number of rooms required.

Makes room planning more efficient.

Improved Allocation Strategy

- A more practical system can use **Best-Fit allocation**.
- **Best-Fit Approach**
- For each subject:
- Find the smallest available room that can accommodate the required students.
- If no single room is sufficient, combine multiple rooms.
- Prefer room combinations with minimum unused capacity.

Example

For **45 students**:

Available rooms:

50 seats

60 seats

80 seats

Best fit = **50-seat room**

Unused seats:

$$50 - 45 = 5$$

This reduces unnecessary empty seats compared with using an 80-seat room.

System Architecture

Input Module

- Subjects
- Student registrations
- Rooms
- Room capacities
- Examination dates and times



Allocation Engine

- Constraint checking
- Greedy / Best-Fit algorithm
- Conflict detection
- Capacity calculation



Validation Module

- Check room capacity
- Check student coverage
- Check scheduling conflicts
- Calculate utilization



Output Module

- Room-wise allocation
- Subject-wise allocation
- Student seating list
- Examination timetable

Advantages and Limitations

Advantages

- Simple and fast allocation.
- Efficient use of available rooms.
- Reduces manual scheduling work.
- Easy to modify when student numbers change.
- Can be extended to include examination dates and time slots.

Possible Advanced Techniques

- Integer Linear Programming
- Constraint Satisfaction
- Genetic Algorithms
- Optimization-based scheduling

Limitations

- Greedy methods do not always produce the mathematically optimal solution.
- Large or complex timetables may require optimization techniques.
- Special requirements such as laboratories or accessibility need additional constraints.
- Student clashes must be considered when scheduling multiple examinations.

Conclusion

- **Key Points**
- College examination room allocation is a **constraint-based scheduling problem**.
- Room capacity and student count are the primary factors.
- A greedy or best-fit strategy provides a practical starting solution.
- Room utilization helps measure allocation efficiency.
- Additional constraints can be incorporated for real-world university requirements.
- **Final Goal**
- **Allocate every student to an appropriate examination room while minimizing wasted capacity and avoiding scheduling conflicts.**